THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE.
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LUDEWA DISTRICT COUNCIL
FORM FOUR MOCK EXAMINATION JUNE 2022.

032/1 CHEMISTRY 1.

(For both school and private candidates)

Time 3 Hours **8th/06/2022**

INSTRUCTIONS

1. This paper consists of section A, B & c with a total of fourteen (14) questions.
2. Answer **all** questions in section A & B and **one (1)** question from section C.
3. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
4. Write your **examination number** on every page of your answer booklet(s).
5. The following constants may be used

   Atomic masses: H=1, O=16, S=32, Ca=40, Zn=65, Cl=35.5, Na=23, C=12, Ag=108

   Avogadro's number = 6.02×10^{23}

   GMV at s.t.p = $22.4dm^3$

   1 faraday = 96,500 coulombs

   1 litre = $1dm^3 = 1000cm^3$

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SECTION A (15Marks)

Answer all questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

i. Mr KALOBERO performed an electrolytic experiment on Calcium chloride.What number of faradays of electricity is required to deposit 4g of calcium from molten calcium chloride?

A. 0.1
B. 0.2
C. 0.4
D. 0.3
E. 0.7

ii. Miss Mfwimi took methane and did substitution reaction with excess chlorine. What is the final product?

A. Chloromethane
B. Dichloromethane
C. Trichloromethane
D. Tetracloromethane
E. Monochloromethane

iii. Mrs. Chakubanga mixed different metals uniformly. Which name is given to the uniform mixture of metals?

A. Compounds
B. Alloys
C. Ores
D. Salts
E. Mixtures

iv. Dr Mbogo did a certain reaction. The reaction was between iodine and hydrogen as represented by the following equations

$$I_{2(g)} + H_{2(g)} \rightleftharpoons 2HI_{(g)} \quad \Delta H = {}^{-}XKJ/mol$$

The reaction is:-

A. An endothermic reaction
B. A displacement reaction
C. A neutralization reaction
D. A thermal decomposition reaction
E. An exothermic reaction

v. A student teacher was in the class teaching how to write electronic configuration of different elements. The electronic configuration of this element Cl^- is

A. 2:8:7
B. 2:8:8
C. 2:8:8:1
D. 2:8:8:2
E. 2:8:6

vi. When Mr Mwendakwao took Glucose and dissolved in water to form a uniform mixture.

A. water is solution, glucose is solvent and the product is solute
B. water is solute, glucose is solvent and the product is solution
C. water is solvent, glucose is solute and the product is solution
D. water and glucose forms immiscible mixture
E. water and glucose form emulsion

vii. Tell to Mr. Kitwanga, left white anhydrous copper (II) Sulphate in air turned blue. The reason why white anhydrous copper (II) Sulphate turns blue when exposed in Atmosphere is that it

A. Absorbs water vapour
B. Reacts with oxygen
C. Reacts with carbon dioxide
D. Become dry
E. Release water to the Atmosphere

viii. In order for Mr. Kingunge to produce the greatest amount of hydrogen in a short time, one gram of magnesium ribbon should react with

A. $10cm^3$ of 0.5M Sulphuric acid
B. $40cm^3$ of 0.5M acetic acid solution
C. $40cm^3$ of 0.5 Sulphuric acid solution
D. $20cm^3$ of 1M Sulphuric acid solution
E. $20cm^3$ of 1M acetic acid solution

ix. Mr. Kigwangala was just complaining about the increases of pollutant gases in the atmosphere causes the increase in temperature. but some gases cause acid were not present to Kigwangala's complain. Make him understand what are the acidic oxides gases.

A. Carbon dioxide, carbon monoxide and Sulphur dioxide.
B. Sulphur trioxide, Nitrogen dioxide and Nitrogen monoxide
C. Carbon dioxide, Sulphur dioxide and dinitrogen oxide
D. Sulphur trioxide, carbon dioxide and Nitrogen dioxide
E. Carbon monoxide, nitrogen oxide and Sulphur dioxide.

x. Mrs.Tupatupa spilled out concentrate acid on the floor. Which action should be taken immediately after concentrated Sulphuric acid is spilled on the skin?

A. It should be rinsed off with large quantities of running water.
B. It should be neutralized with concentrated NaOH
C. The affected area should be wrapped tightly and shown to a medical health provider
D. It should be Neutralize with solid $CaCO_3$
E. It should be neutralized with concentrated KOH

2 Match the items in List A with the response in List B by writing the letter of the corresponding responses beside the item number.

LIST A	**LIST B**
i. Biogas	A Industrial gas which is a mixture of hydrogen and carbon monoxide
ii. Biomass	B Non- renewable gaseous fuel
iii. Natural gas	C Industrial gases fuel which is a mixture of nitrogen gas, hydrogen and carbon monoxide
iv. Producer gas	D Gases fuel derived from decomposition of biological waste
v. Water gas	E Renewable source of energy
	F Water gas and ammonia
	G Ammonia and butane

SECTION B (70Marks)

Answer all questions in this section

3. (a)(i) A student from pre form one class asked you teach him the real meaning of Laboratory. What was you explained so as to make him understand the meaning of laboratory?

(ii) Mr. Kiseyeye wanted to prepare an oxygen. What are two main compound do Mr. Kiseyeye used in preparation oxygen gas in laboratory.

(ii)You as a student from form four class what do you think Why hydrogen gas is collected by downward displacement air method?

(b)Mr Kadinya don't understand how to complete and balance chemical reactions. Help mr Kadinya to Complete and balance the following chemical reaction

i. $NaOH_{(ag)} + HCl_{(ag)} \longrightarrow$
ii. $NH_4Cl_{(ag)} + NaOH_{(ag)} \longrightarrow$
iii. $Ca(OH)_{2(ag)} + H_2SO_{4(ag)} \longrightarrow$

iv. $Mg_{(s)} + HNO_{3(ag)} \longrightarrow$

4. Element R having atomic number 20 combines with element S having atomic number 17 to form a certain compound
   a) Write the formula of the compound and state the type of bond formed in the compound
   b) Give any three properties of the compound formed in 7(a) above
5. .As a competent chemistry student chosen to present the topic of mole concept and related calculations. Show how will you present the following calculations
   i. How many chlorine molecules are in $20cm^3$ of chlorine gas at s.t.p.

   ii) Calculate number of ions present in 5g of copper (II) Nitrate
6. a) Muumini is a student from Katosani secondary school, she is not able to State Le Chatellier's principle, Help her.

   b) In the industrial preparation of Sulphur trioxide mr Atieno yo established equilibrium between Sulphur dioxide and oxygen gas as follows:

   $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} \Delta H = -94.9 KJ/mol.$

   i. How would you adjust temperature and pressure to maximize the proportion of the product at equilibrium?
   ii. Why is it unfavorable to work with very high pressure and very low temperature in the contact process?
   iii. What catalyst is used to speed up the rate of formation of Sulphur trioxide before attaining the equilibrium?
7. a) You are tasked to explain the meaning of Molecular formula in a certain presentation that will be held next year. Present it now.
   b) You are provided with a compound composed of 22.2% Zinc, 11.6% Sulphur, 22.3% oxygen and the rest percentage is water of crystallization. Calculate the molecular formula of the compound if its molecular mass is 283.
8. In one of titration experiment $25cm^3$ of hydrated sodium carbonate $Na_2CO_3.XH_2O$ were titrated against $20cm^3$ of 0.25M hydrochloric acid. The solution of sodium carbonate was made by dissolving 7.15g of the compound in $250cm^3$ of distilled water.
   Use the above data to answer the following questions
   a) Write down a balanced equation to represent the acid base reaction
   b) i. Find the morality of the an hydrated sodium carbonate
      ii. Calculate the value of X in the hydrated sodium carbonate.
9. a) Mr Mapulishi tasked to prepare all requirements for extraction of sodium.Describe the use of each of the following during extraction of Sodium
      i. Calcium Chloride
      ii. Graphite anode
      iii. Steel gauze
   b) .Why sodium is collected by upwards in the downs cell
   c) Write electrodes reaction in downs cell during extraction of Sodium
10. a) When I was in an interview at Feza boys' secondary school, one student asked me to State Faraday's Laws of Electrolysis. If you were me answer such question.
    b) Dilute silver nitrate solution was decomposed by the passage of electric current through it. What mass of silver and what volume of oxygen (Measured at s.t.p) would be liberated in electrolysis by 9650 coulombs of Electricity?
11. Mr Kalubandika wanted to know some chemistry pertaining concepts. Help Mr Kalubandika to answer the following conceptual questions.

a) In which other areas do we find the warning signs out of laboratory (give four point)

b) Explain how measurements of volume differ when using measuring cylinder and burette

c) It is recommend that laboratory apparatus should be properly washed or wiped after use, explain the significance for this when

i. Measuring volume of liquids

ii. Measuring mass of substance

12. a) Mr Mwakatumbula don't understand the physical properties of water. Teach him by giving three points the main physical properties of water and show the usefulness of each property.

b) Mr Kadinya said that Electrolysis is applied by many areas on the earth. State four industrial application of electrolysis

SECTION C (15 MARKS)

13. We have coal at Kiwira in Mbeya region. Authorities in the government have allowed use of coal for domestic and industrial purpose. What warming can you raise concerning likely effects? Give five points

14. a). Explain the mechanism of electrolysis

b). briefly explain three application of electrolysis in our daily life